Hybrid Spatiotemporal Graph Convolutional Network for Detecting Landscape Pattern Evolution From Long-Term Remote Sensing Images

Meiling Liu, Xiangnan Liu, Ling Wu, Tao Peng, Qian Zhang, Xinyu Zou, Lingwen Tian, and Xuan Wang

***Abstract*— The remote sensing time series change detection algorithm based on the pixel or single landscape patch ignores the change analysis of spatial structure information. Inspired by graph convolutional network (GCN) modeling, a set of landscapes (nodes) and their relationships (edges) are proposed. In this study, a parallel strategy in spatial-GCN and progressive strategy in temporal-GCN, called the hybrid GCN model network as a holistic framework, was proposed to accurately capture both spatial and temporal variations in landscape patterns based on yearly Landsat time series. A superpatch (i.e., fixed patch class surrounded by a one-hop neighbor patch) was selected as the input of the GCN network. First, a spatial-GCN model with three parallel graph convolutional layers was adopted to classify landscape pattern types. Three dominant categories of landscape patterns over the past three decades have been identified. Second, four landscape metrics in superpatches were proposed for the quantitative characterization of changes in landscape patterns. Finally, a temporal-GCN model with two progressive graph convolutional layers was used to detect six types of patch changes, which were applied to continuously detect the landscape pattern evolution processes. Regardless of the spatial-GCN and temporal-GCN, they provided satisfactory performance using the training and validation sets with overall accuracy >92% and Kappa coefficient >0.90, and loss values converging to 0.049 and 0.128, respectively, based on NLLLoss function until 500 epochs. It is believed that the hybrid GCN model has great potential for mining possible implicit spatiotemporal relationships and future evolution of landscape patterns.**

***Index Terms*— Graph convolutional network (GCN), Landsat time series, landscape pattern, superpatch spatial graph.**

I. Introduction

LANDSCAPE patterns are referred to as the spatial configuration of various landscape elements and their compositions, which influence the ecological processes that lead to biodiversity, habitat quality, ecological functions, and ecological security on local-to-global scales [1], [2], [3]. With the establishment of the U.S. Geological Service, the Landsat data archive provided us with a more comprehensive understanding of how the Earth's surface dynamics will react to natural and human influences, and it offers a good opportunity to measure detailed and consecutive landscape patterns with a spatial resolution of 30 m and longest running historical archive [4]. Changes in landscape patterns are primarily related to land-use conversion and modification, particularly those induced by anthropogenic disturbances [3]. Mapping landscape pattern changes and understanding the underlying processes are priorities for the major design target of low-medium spatial remote images (e.g., Landsat images) but have also been an important technology approach in geography and landscape ecology research [5]. It is more strongly advocated as underpinning future landscape pattern dynamics, sustainable and optimized ecological patterns, and assessments of the life community with "hill, lake, field, and grass."

Many researchers have attempted to determine the temporal–spatial dynamics of landscape patterns in the study area [3], [6], [7], [8]. The temporal–spatial dynamics of landscape patterns according to the analysis unit can be categorized into landscape cells and patches. Landscape cells obtain landscape metrics by using a moving window or kernel method of a fixed size, shape, and (limited) orientation(s) of the regular image pixels. Compared with landscape cells, landscape patches represent "meaningful" land cover class or scene components that are distinguishable in an image, which capture a combination of objects such as shape, size, context, and pattern [9]. Landscape patches play a useful role in comparing spatial arrangements, for example, the landscapes of different areas at the same time and the landscapes of the same area at different times [10], [11]. Therefore, landscape patches have been widely regarded as effective avenues for land-use dynamics and landscape pattern change processes [12], [13], [14].

Recently, an important trend has been the use of landscape patches to handle landscape pattern changes using satellite image time series data. Tian *et al.* [12] obtained landscape patches by the classification of land use or land cover and

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then measured the dynamic changes in cropland patches, such as perforation, dissection, shrinkage, creation, enlargement, and aggregation types using morphological-rule-based pattern recognition. Several studies have focused on detecting the spatiotemporal dynamics of different land cover patches derived from image segments [13], [14]. Significant progress has been made in this regard. However, these studies mainly considered the evolution behavior of patches in their own state (such as the change in area, perimeter, and shape) and focused on the analysis of the temporal characteristics of a single patch. The evolution behavior of the relationship between the landscape patch itself and the spatial adjacency patch (such as spatial structure and distribution) was ignored, which resulted in the separation of the temporal and spatial relationships of multipatches.

Based on the above analysis, there are multiple challenges in applying remote sensing to detect landscape patterns in the study area. 1) patch–patch interaction, such as the interaction among forest, cropland, and building patches in landscape design planning; 2) process–pattern feedback, such as landscape pattern changes as a result of a sequence of many small patch abrupt changes; and 3) high-level spatiotemporal features, which are the major change categories of landscape patterns in the whole study area over a long period. Solving these problems can help to analyze landscape pattern evolution processes and are more strongly advocated as the underlying reasons and mechanisms of landscape pattern evolution.

The method based on landscape patches is more attractive and proves meaningful in the pattern–process dynamic analysis. To extract and analyze the evolution of landscape patterns and their temporal and spatial relationships in long-term remote sensing images, priority should be given to selecting an effective model to characterize the complex temporal and spatial relationships of landscape patterns.

Previous studies have proven that graph models have advantages in mining the landscape patch relationships for complex landscape ecosystems based on the strength of their interconnections presented in spatiotemporal data models [15], [16]. To date, the most promising and widespread approach for modeling landscape patches has been based on graph models. Cantwell and Forman [17] demonstrated that the centroids of landscape patches are conceptualized as nodes (vertices). The connections among nodes, called links (linkages and edges), generally represent the distance between nodes in landscape ecological studies. Therefore, it is feasible to analyze landscape dynamics by focusing on patch–patch interactions through the connection of the edges and nodes of these patches. Cheung *et al.* applied a graph model to support spatiotemporal analysis in landscape monitoring and to detect landcover changes [16], [18]. With the development of machine learning technologies, graph convolutional networks (GCNs) have gained attention in the research community because of their ability to competently process irregular image regions in a non-Euclidean (or nongrid) graph data structure [19].

We propose a hybrid GCN as a holistic framework for efficiently capturing both spatial and temporal changes in landscape patterns. Our model addresses three technical difficulties: 1) patch–patch localized spatial dependency; 2) temporal dynamics in interactive landscape patches; and 3) characterization of the spatiotemporal evolution of landscape pattern types in the study areas.

II. Related Work

A. Change Detection Algorithm

The information levels of changes from remote sensing imagery can be classified into the following.

1) Pixel-based change detection methods, which focus on exploiting the spectral characteristics of an image pixel to detect and measure changes. These techniques include postclassification comparison, image differencing [20], change vector analysis [21], and long time series analysis, such as Landsat-based Detection of Trends in Disturbance and Recovery (or LandTrendr) [22], Vegetation Change Tracker (VCT) [23], and Breaks for Additive Season and Trend (BFAST) [24].
2) Spatial context change detection methods, which are similar to the pixel-based changes in the analysis of the nature of the change; spatial context refers to the spatial neighborhood of a focal pixel (the pixel that is being processed). However, it allows the full use of the spatial information of the current pixel, such as texture and spatial relationships by using the spatial moving window [25], [26].
3) Object-based change detection, which refers to identifying differences in geographic objects at different times by object-based analysis.

Tian *et al.* [12] developed morphological-rule-based pattern recognition and quantified various farmland patterns based on state-and-evolution detection models using yearly Landsat data from 1988 to 2018. Khiali *et al.* [14] detected the spatiotemporal dynamics of geographic entities by means of a graph structure using a multiannual satellite image time series. Guttler *et al.* [13] extracted the spatiotemporal evolution of graph and study site levels based on the graph structure using a combination of data mining techniques and global variation parameters. The above change detection approaches focused their efforts on detecting the spatiotemporal dynamics of a single pix or landscape patch. However, they do not provide any information regarding the interaction change of multipatches, which helps to interpret the cause of changes (i.e., how and why?). To cope with these issues, a superpatch, which is referred to as a fixed patch class surrounding all patches in a one-hop neighbor patch, was selected as a unit analysis, and we focused on the change in patch–patch interactions and spatial–temporal relationships.

B. Graph Convolutional Network

GCNs generalize convolutions to graphs of arbitrary structures and aim to learn low-dimensional latent representations of graphs. It is a useful method for processing non-Euclidean data. GCN has many applications in various aspects: node classification (such as social network node classification), hyperspectral image classification [27], [28], link prediction (such as traffic forecasting and traffic accident prediction) [29], and visualization (such as computer vision) [30]. Significant

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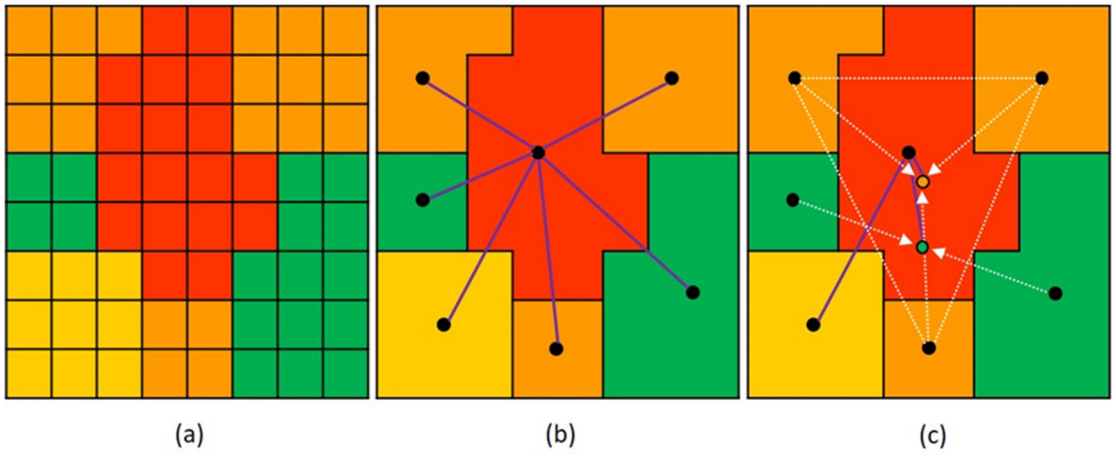

Fig. 1. Land cover spatial graph. (a) Landscape snapshot, note that the same color represents the same land cover type. (b) Spatial graph based on land cover patch, note that the black dots represent the geometric centroid of each land cover patch. (c) Spatial graph based on land cover types, note that green dot represents the centroid of line segment consisting of the black dot for the geometric centroid of the two green land cover patch and orange dot represents the centroid of triangle consisting of the black dot for the geometric centroid of the three orange land cover patch.

progress has been made in node classification and edge prediction by applying GCN. However, the use of GCN to detect landscape patterns has rarely been addressed. GCN has promising potential for extracting valuable knowledge, namely, the patch–patch spatial dependency relationships, possible implicit relationships between the landscape patch itself and its neighbor patch through a neural network, and higher level feature representations by gradually aggregating low-level features. Therefore, GCN can exhaustively exploit the evolution of landscape patterns.

Notably, the present GCN model mainly focuses on the analysis of the static graph, which deals with the properties of nodes and edges that remain unchanged over time. Landscape patterns present a great challenge because the graph may have a variable size of nodes due to patches appearing or disappearing in a particular period, and nodes from a graph can differentiate between periods using the number of neighbors, which makes graphs dynamic. Therefore, the proposed GCN can support dynamic spatiotemporal relational analyses for landscape pattern monitoring.

III. Proposed Method

A. Construction of Graph Structure

A graph can be defined as a data structure composed of a set of nodes and edges and is used to describe the relationship between different nodes [31], [32]. The features of nodes and edges are important for establishing the proposed spatiotemporal graph in the model. The construction of spatial and temporal is described.

1) Spatial Graph: First, in this study, different land cover types were derived from the images [Fig. 1(a)]. Second, each node represents the geometric centroid of its corresponding land cover patch. If two land cover patches share a boundary, an edge is generated on their corresponding node [33], [34]. In addition, the graph is directed when the edges are ordered pairs from central nodes, which are selected based on the fixed land cover types (such as cropland) to neighbor nodes, where the number of edges within a one-hop neighbor landscape unit is equal to the number of patches [Fig. 1(b)], and note that a one-hop neighbor refers to the object that is located in the vicinity of a certain routing path between source and destination [35], [36], that is to say, a one-hop neighbor landscape unit is defined as the landscape patch that is located in the vicinity of central landscape patch (fixed landscape patch) with overlap boundary between the central landscape patch and the neighbor patch. Third, the different patches belong to the same land cover type within the one-hop neighbors' landscape unit, where each node represents the geometric centroid of its corresponding polygon or line segment consisting of the geometric centroid of each land cover patch [Fig. 1(c)].

2) Temporal Graph: The node in the temporal graph is generated based on the land cover patch, which is the same as the node in the spatial graph [Fig. 1 (b)]. The time edge is referred to as the temporal connection between the two objects that reside (Fig. 2). The time edge between two land cover patches is established if they overlap in two adjacent snapshots of the landscape [16], [37], for example, there is overlapping in the land cover patch (A' and $C1'$) in the previous snapshot at time $t-1$ and the land cover patch (A) in the current snapshot at time t [Fig. 2(c)].

B. Parameters for Characterizing Landscape Pattern

The landscape pattern changes were classified into two ecological processes: landscape patch type change (such as farmland converted forest) or landscape patch movement (such as the change in patch area and shape). The former was identified by the variation of node attributes, and the latter was identified by the variation of distance and direction, which can be advocated as underpinning configuration of the landscape pattern; therefore, node attributes, distance, and direction were used to characterize landscape pattern types. The details are given as follows.

1) Node (Patch) Attribute: Landscape patches are homogeneous geographical units. Here, each patch was identified

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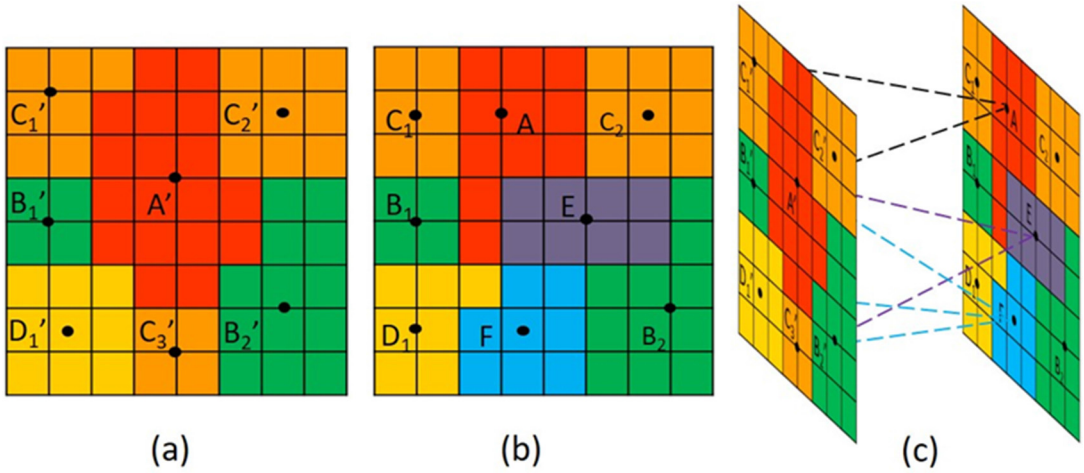

Fig. 2. Land cover temporal graph. (a) Landscape snapshot at time $t-1$, note that the same color represents the same land cover type. (b) Landscape snapshot at time t. (c) Time edge in two adjacent snapshots of the landscape, for example, land cover patch A, E, and F at time t have two, two, and three time edges with black, purple, and cyan line, respectively.

according to land cover classes, namely, cropland, forest, grassland, water, building land, and unused land, to identify different landscape patterns, which are constructed as follows:

$$\forall x_i, x_j, x_k, x_m, x_n, x_l \in \{x_1, x_2, x_3, x_4, x_5, x_6\},$$
$$\text{in addition, } x_i \neq x_j \neq x_k \neq x_m \neq x_n \neq x_l$$
$$\text{LPT} = \begin{cases} (x_i, x_j), \\ \quad \text{if land cover type number in neighbour node} = 1 \\ (x_i, x_j, x_k), \\ \quad \text{if land cover type number in neighbour node} = 2 \\ (x_i, x_j, x_k, x_m), \\ \quad \text{if land cover type number in neighbour node} = 3 \\ (x_i, x_j, x_k, x_m, x_n), \\ \quad \text{if land cover type number in neighbour node} = 4 \\ (x_i, x_j, x_k, x_m, x_n, x_l), \\ \quad \text{if land cover type number in neighbour node} = 5 \end{cases} \tag{1}$$

where LPT denotes landscape pattern groups based on the landscape patch types, x_i is the central node, x_j, x_k, x_m, x_n, x_l are neighbor nodes, and $x_1, x_2, x_3, x_4, x_5, x_6$ are cropland, forest, grassland, water, building land, and unused land, respectively. A sketch map of the landscape pattern groups based on landscape patch types is shown in Fig. 3(a).

2) Distance and Direction of Patch–Patch: The distance and direction were confirmed as the distance and angle, respectively. Here, we obtained the principal axis; the angle between the central node and neighbor nodes rotates to a degree in a clockwise direction to achieve a variant spatial signature, assuming the eastern direction as the original direction. Let (a_1, a_2) and (b_1, b_2) be the locations of the centroids of the central node and neighbor nodes, respectively, and let θ_1 and θ_2 be the angles of neighbor node A and neighbor node B, respectively. The distance (d) and the angle of the pairwise nodes (adjacent nodes) (θ) are given as follows:

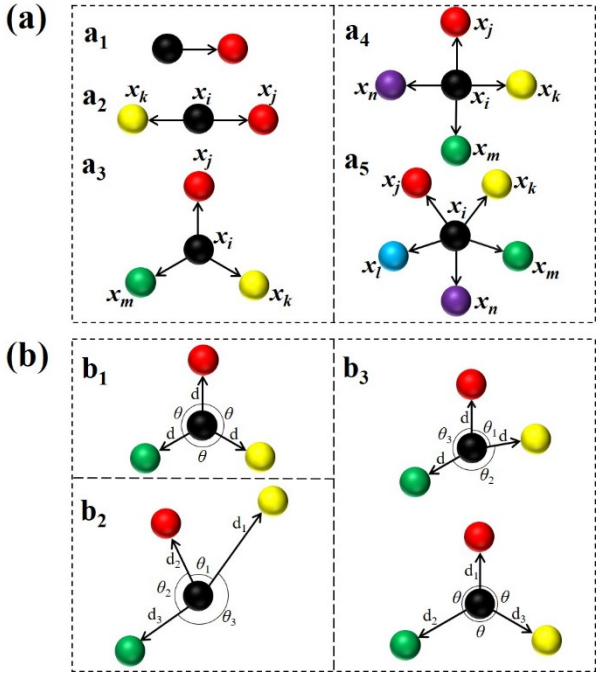

Fig. 3. Sketch map of the landscape pattern. (a) Possible landscape pattern groups based on the patch relating to land cover types. (b) three landscape pattern types based on the direction and distance as an example for land cover type number in neighbor node equaling four; b_1 denotes RUP, b_2 denotes IUP, and b_3 denotes SUP.

$$d = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2} \tag{2}$$

$$\theta = \begin{cases} |\theta_1 - \theta_2|, & \text{if } |\theta_1 - \theta_2| \leq 180 \\ 360 - |\theta_1 - \theta_2|, & \text{if } |\theta_1 - \theta_2| > 180. \end{cases} \tag{3}$$

Based on the distance (d) and direction (θ) between the central and neighbor nodes, we defined a spatial signature

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for measuring the landscape pattern, which was classified into three categories [Fig. 3(b)], namely, relatively uniform pattern (RUP), incompletely unbalanced pattern (IUP), and severely unbalanced pattern (SUP).

a) RUP: This pattern comprised regions characterized by the same distance and angle within the landscape unit of one-hop neighbors [Fig. 3(b1)]. The patterns showed regular and similar-sized patches and the lowest degree of fragmentation.

b) SUP: This pattern was characterized by different distances and angles [Fig. 3(b2)]. The SUP was characterized by a number of patches of different shapes and sizes, and a few patches of the same land cover type with small areas.

c) IUP: The deviation degree of the distance and angle for this pattern is between that of RUP and SUP. With regard to a similar distance, IUP indicates that the larger central patches are highly rounded with a few of the same neighbor patches. With regard to a similar angle, the IUP was mainly defined by a relatively small central patch and larger neighbor patches.

The four most important indicators, namely, central node type, neighbor node type, distance, and direction between the central and neighbor nodes, which influence the landscape pattern types, are considered. Therefore, in theory, all available landscape pattern groups were calculated based on mathematical permutations, and the number of landscape pattern types is expressed as follows:

$$\mathrm{LP}_n = C_{dn}^1 \times \mathrm{LPT} = C_{dn}^1 \times C_{Tn}^1 \times \left(C_{Tn-1}^1 + C_{Tn-1}^2 + \cdots C_{Tn-1}^{Tn-1}\right). \quad (4)$$

Here, dn is 3, which denotes three landscape pattern types (i.e., RUP, IUP, and SUP), on the direction and distance, Tn is 6, which denotes the land cover class number (i.e., cropland, forest, grassland, water, building land, and unused land). C_{Tn}^1 (i.e., C_6^1) represents the available group of central node from six land cover classes.

C_{Tn-1}^1 (i.e., C_5^1), C_{Tn-1}^2 (i.e., C_5^2), and C_{Tn-1}^{Tn-1} (i.e., C_5^5) are the available group when land cover type numbers in the neighbor node are 1, 2, and 5, respectively. In theory, 558 landscape pattern groups for six land cover class variables were distributed in the study area. The landscape pattern groups were equal to the number of landscape patches in the study area.

C. Establishment of Hybrid Spatiotemporal GCN

Throughout this article, bold uppercase and lowercase characters denote matrices and vectors, respectively, unless specified. A graph can be described with formula $G = (V, E, A)$. V denotes the node set of the graph and E is the edge set of the graph. $A \in \mathbb{R}^{N\times N}$ denotes the adjacency matrix. $X \in \mathbb{R}^{N\times M}$ denotes a feature matrix of N nodes with M-dimensional features in a graph. Edge attribute $X^e \in \mathbb{R}^{Q\times C}$ represents a feature matrix in Q edges with C-dimensional features in a graph.

The GCN unit takes the feature matrix X as the input and conducts the spectral graph convolution operation (according to A) with a localized first-order approximation. An adjacency matrix of graph G with added inserted self-loops is expressed as follows:

$$\hat{A} = A + I \quad (5)$$

where I is the N-dimensional identity matrix. The elements in the diagonal degree matrix $\widehat{D}_{ii}$ were calculated as follows:

$$\widehat{D}_{ii} = \sum_{j=0} \hat{A}_{ij} \quad (6)$$

where A_{ij} represents the element in the ith row and jth column position of the adjacency matrix.

The calculation process for the state update in a GCN unit can be defined as follows:

$$f\left(H^{(l)}, A\right) = \sigma\left(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right) \quad (7)$$

where $f(H^{(l)}, A)$ is the output, $H^{(l)}$ is the activation of the lth layer, $\widehat{D}^{-(1/2)}\hat{A}\widehat{D}^{-(1/2)}$ with $\hat{A}$ and $\widehat{D}$ the normalized matrix, $W^{(l)}$ represents the learnable weight parameter in this lth layer, and $\sigma(\bullet)$ represents an activation function, such as ReLU($\bullet$)

In this study, a spatiotemporal GCN focused on the relationship between time and space; therefore, the spatial- and temporal-GCN models are given as follows.

1) Spatial-GCN Model: The spatial-GCN model aims at capturing the spatial distribution of landscape pattern types and its major types. The model helps to distinguish the dominant landscape types in time t. The model consists of input data (i.e., node feature, edge feature, and node adjacent matrix) and output data (i.e., landscape pattern types). The chart flow of the spatial-GCN model is shown in Fig. 4. From Fig. 4, we can see that a graph convolution layer includes two steps.

1) Generating three new feature representations from the input $Z^{(l)}$ by performing three parallel graph convolution layers; the first and second layers were used to extract the North–South and East–West offsets relations of the central node and adjacent nodes, respectively, by input different edge weight ($e_{ij}^{\prime(x)}$ and $e_{ij}^{\prime(y)}$), and the third layer was to extract the distance between central and adjacent nodes.
2) Feeding the new generated feature to a multilayer fully connected layer to obtain the node category information from the spatial relationship between nodes to characterize the landscape pattern types. In addition, an activation function ReLU was adopted in three parallel graph convolution layers and fully connected layers.

The detail feature matrix X and the adjacent matrix A were obtained as follows.

The feature matrix X includes land class types and node degree, namely

$$X_i = \left\{x_1, x_2, x_3, x_4, x_5, x_6, du'\right\}, \quad X_i \in X \quad (8)$$

$$du' = \frac{du}{\max(du)}. \quad (9)$$

Here, x_1, x_2, x_3, x_4, x_5, and x_6 are cropland, forest, grassland, water, building land, and unused land, respectively. One-hot encoding and N-bit status registers were used to encode N states, and each state had its own independent register

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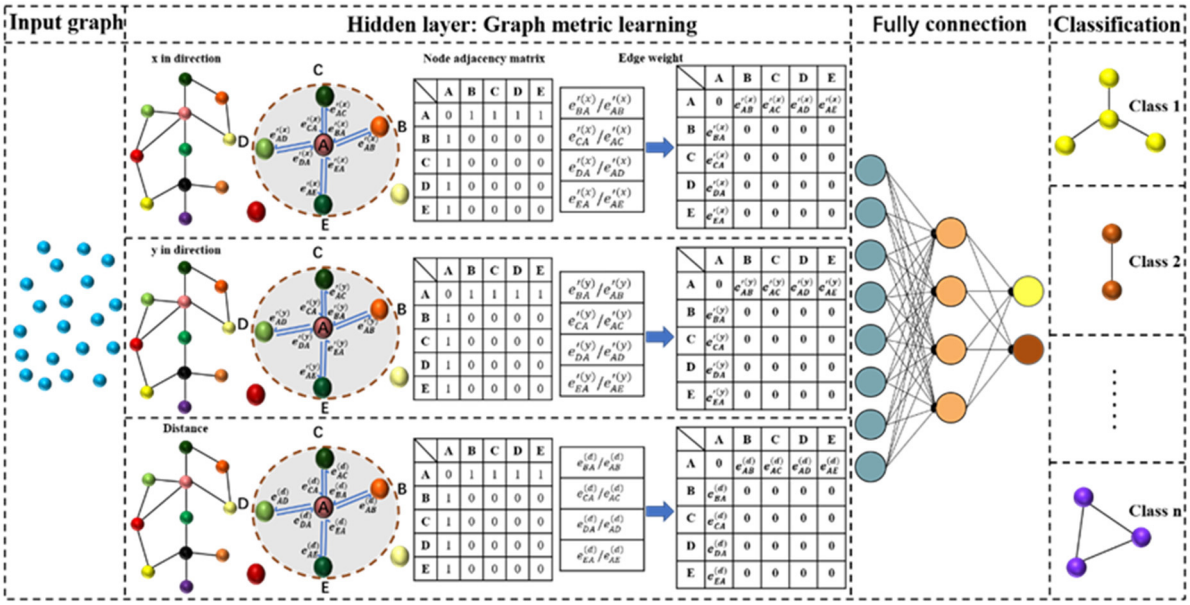

Fig. 4. Classification of landscape pattern types based on spatial-GCN models.

bits [40], which was used to memorize node features correlating to land class types. du is the number of central nodes connected with neighbor's node land class types; du' is the normalization of du, which ranges from 0 to 1; $\max(du)$ is the maximum of the central node connected to the neighbor's node land class types in the study area. Here, $\max(du)$ is five.

The adjacent matrix is calculated as follows:

$$A_S = A * X_s^e \tag{10}$$

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \in \text{Nei}(v_j) \text{ or } v_j \in \text{Nei}(v_i) \\ 0, & \text{otherwise} \end{cases} \quad a_{ij} \in A. \tag{11}$$

Here, $\text{Nei}(v_j)$ and $\text{Nei}(v_i)$ are all neighbor's node of v_j and v_i, respectively. A is the original adjacency matrix without weight value. A_s is the adjacency matrix after assigning spatial weights, which was taken as the final form of the adjacent matrix in our model. X_s^e is the weight matrix of edge, and it includes the x-axis weights ($e_{ij}^{\prime(x)}$), the y-axis directions weights ($e_{ij}^{\prime(y)}$), and distance weights ($e_{ij}^{(d)}$). The detail was obtained as follows:

$$e_{ij}^{\prime(x)} = \frac{e_{ij}^{(x)} - \min\left(e_{ij}^{(x)}\right)}{\max\left(e_{ij}^{(x)}\right) - \min\left(e_{ij}^{(x)}\right)}, \quad \text{where } e_{ij}^{(x)} = \cos(\theta) \tag{12}$$

$$e_{ij}^{\prime(y)} = \frac{e_{ij}^{(y)} - \min\left(e_{ij}^{(y)}\right)}{\max\left(e_{ij}^{(y)}\right) - \min\left(e_{ij}^{(y)}\right)}, \quad \text{where } e_{ij}^{(y)} = \sin(\theta) \tag{13}$$

$$e_{ij}^{(d)} = \frac{d_{ij}}{\max(d)}. \tag{14}$$

Here, the same weights among all spatial edges are based on the direction and distance within the landscape unit of one-hop neighbors (Fig. 5). $e_{ij}^{(x)}$ and $e_{ij}^{(y)}$ denote the x- and y-axis directions, respectively, between the central and neighbor nodes [Fig. 5(a)]. $e_{ij}^{(d)}$ denotes the distance between the central and neighbor nodes [Fig. 5(b)]. Notably, the e_{ij} weight

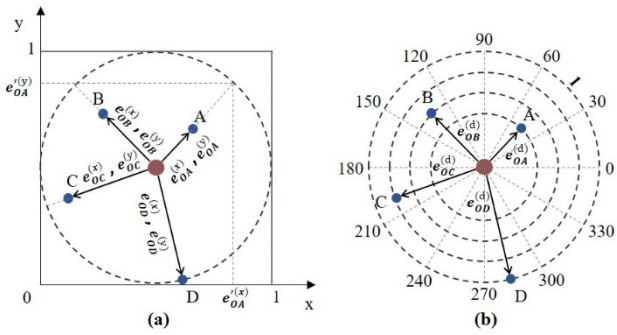

Fig. 5. Spatial configuration between central and neighbor nodes. (a) Angle and direction. (b) Distance.

of the edge had its own independent value in three parallel layers of the GCN.

2) Temporal-GCN Model: The temporal-GCN model monitors temporal changes in landscape pattern types by detecting changes in single patches. By applying the temporal-GCN model, five possible temporal changes can be identified for a single patch: creation, aggregation, dissection, enlargement, and shrinkage [12]. The creation, aggregation, and dissection can lead to a patch-type change, whereas enlargement and shrinkage result in the patch area, which makes the distance and direction between patch and neighbor patch. Regardless of the five possible temporal changes in a single patch, the change in landscape pattern occurred with the change in a single patch. Singe patch change coexists with landscape pattern change; the former provides the reason and mechanism for the latter.

In this study, the temporal-GCN model consisted of input data (i.e., node features, edge features, and node adjacent matrix) and output data (i.e., change types). The flowchart of the temporal-GCN model is shown in Fig. 6. From Fig. 6, suppose that node A is regarded as the central node

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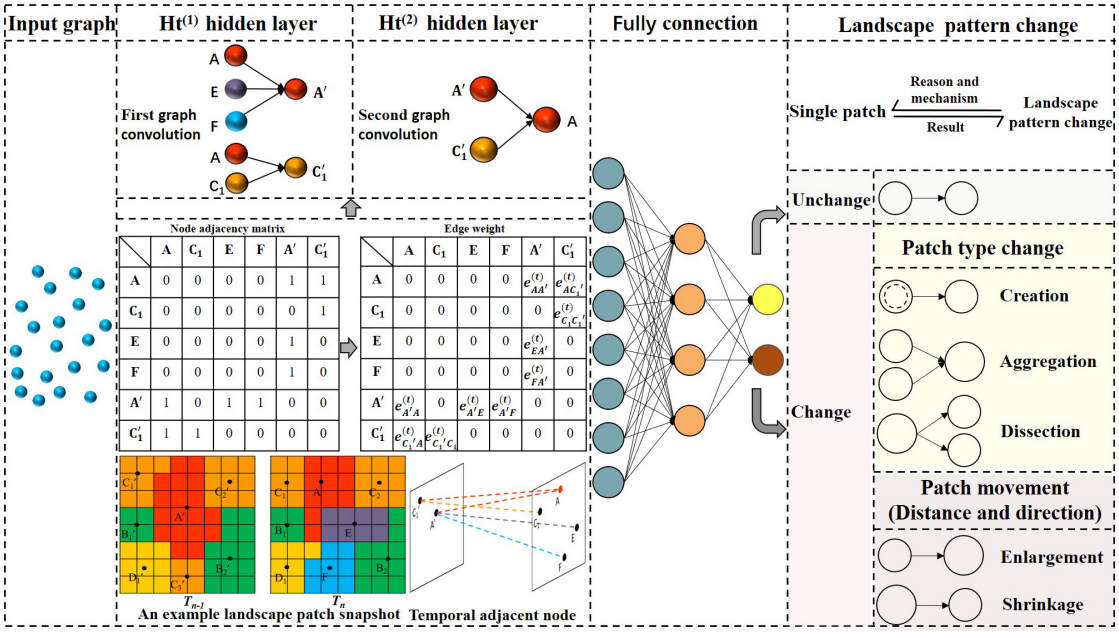

Fig. 6. Dynamic change in landscape patterns through the change of a single patch based on temporal-GCN models.

(i.e., the detected node). The change in node A was analyzed in the following steps.

1) Generating new feature representations from the input data by performing two successive graph convolutions; the first graph convolution layer was used to extract the relationship between the two- and one-hop temporal adjacent nodes of the central node ($H_t^{(1)}$ hidden layer). The second graph convolution layer was used to obtain the direct relation between the central and one-hop temporal adjacent nodes by obtaining the comprehensive features (original features and relational features given by the first layer ($H_t^{(2)}$ hidden layer).
2) Feeding the newly generated feature to a multilayer fully connected layer to obtain the node change category information from the temporal relationship between nodes to characterize the change in the landscape pattern. In addition, an activation function ReLU was adopted in two successive graph convolutions and fully connected layers.

The time edge weight ($e_{ij}^{(t)}$) was calculated as follows.

The temporal connection between two patches is established if the corresponding objects overlap in two successive timestamps. Two nodes are linked by an edge in two successive timestamps, which is referred to as a time edge. The time edge weight was determined by the ratio of the overlapping area to the area of the patch. Namely,

$$e_{ij}^{(t)} = \begin{cases} \dfrac{\left(\mathrm{So}_i^t \cap \mathrm{So}_j^{t-1}\right)}{\mathrm{So}_i^t}, & \text{if temporally connected from } t-1 \text{ to } t \\ \dfrac{\left(\mathrm{So}_i^t \cap \mathrm{So}_j^{t-1}\right)}{\mathrm{So}_j^{t-1}}, & \text{if temporally connected from } t \text{ to } t-1 \end{cases} \tag{15}$$

$\left(\mathrm{So}_i^t \cap \mathrm{So}_j^{t-1}\right)$ denotes the overlapping area between the two patches in the two snapshots (at time $t-1$ and t), which is calculated by intersecting the geometric shape of the two patches. Notably, $e_{ij}^{(t)}$ is a directed edge. As indicated by the criterion below, if the temporal connection of the two patches is directed from $t-1$ to t, the patch area at time t (So_i^t) is considered the denominator of the ratio. Conversely, if the temporal connection of the two patches is directed from t to $t-1$, the area of the patch in time $t\mathrm{So}_j^{t-1}$ is considered as the denominator of the ratio.

D. Analysis of Spatiotemporal Variation of Landscape Patterns

To effectively detect spatiotemporal variations in landscape patterns, multiple aspects may be optimal for monitoring landscape patterns in the study area.

1) Evolution Time: To execute the landscape pattern time step, the evolution time of each type is assigned an initial time at the first detected state type, and this time is then updated every time step until the next different landscape pattern type occurs.

2) Evolution Metric: The four evolution metrics in the one-hop neighbors' patch were selected, as detailed in Table I and Fig. 7. Patch connectivity (PC), distance of the two adjacent patches (DP) largest patch index (LPI), and perimeter–area fraction (PAF) represent connectivity, containing, domination, and complexity of the superpatch, respectively, thereby characterizing the dynamics of the spatial arrangement of the surface landscape.

IV. Experiments and Analysis

A. Experimental Datasets

The study area (112°17.7′E–112°20.9′E and 37°55.1′N–37°58.3′N) was located in Taiyuan, Shanxi, China, where

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TABLE I
FOUR EVOLUTION METRICS IN ONE-HOP NEIGHBORS' PATCH USED IN THIS STUDY

Metrics	Definition	Acronym	Formula	Notes
Patch connectivity	number of central patches directly connected to the neighbor patch	PC	–	See Fig.7 （a）
Distance of the two adjacent patches	total sum of the distance from the geometric centroid of the two patches in all neighbor patches	DP	$DP=\sum_{i=1}^{n} d_i$	See Fig. 7 （a）
Largest patch index	the maximum area of the patch divided by the average size of all patches contained in one-hop neighbors' patch	LPI	$LPI=\frac{\max(S_j)}{\frac{\sum_{j=0}^{n} S_j}{n+1}}$	See Fig. 7 （b）
Perimeter- area fraction	the ratio of central patch perimeter and central patch area in one-hop neighbors' patch	PAF	$PAF = 0.25\,P/\sqrt{S_0}$	See Fig. 7 （b）

Notes: n, P, S_0 and S_j , represent the number of neighbor patches, perimeter of the central patch, central patch area and area of some onE-HOP NEIGHBORS' PATCHES, RESPECTIVELY (FIG.7).

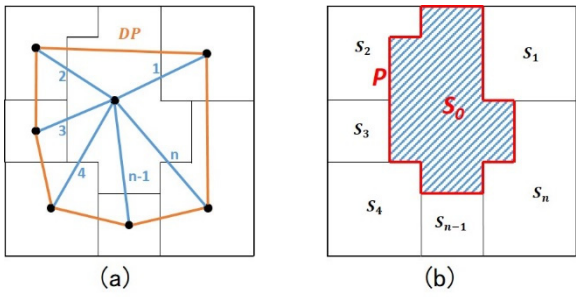

Fig. 7. Sketch map for four evolution metrics in one-hop neighbors' patch in this study. (a) PC and DP. (b) LPI and PAF. Note: description about n, P, S_0, and S_j is the same as in Table I.

the remarkable and rapid landscape pattern changes occurred due to the land surface of Shanxi undergoing frequent modification [37]. Landsat images derived from Landsat-5 TM during 1987–2000 and 2003–2012, Landsat7-ETM + during 2001–2002, and Landsat-8 OLI during 2013–2020 and confined to the acquisition period between June and September for each year were collected. In this study, a random forest classifier deployed on Google Earth Engine (GEE) was adopted to obtain six land cover classes, namely, cropland (labeled 1), forest (labeled 2), grassland (labeled 3), water (labeled 4), building land (labeled 5), and unused land (labeled 6), using all spectral bands of images as input parameters. Elevation and slope were calculated using the GEE digital elevation model to assist with distinguishing forest from grassland and cropland. The final land cover maps comprised six land cover classes: cropland, forest, grassland, water, building land, and unused land, and the reported average overall accuracies (OAs) over 35 years were 80.19%, 92.56%, 77.06%, 82.18%, 89.04%, and 97.63%.

B. Experimental Settings

In this study, all experiments were conducted on a personal computer with the following hardware: Intel Xeon CPU E5-2620 V4, NVIDIA RTX 2080 graphics card, and 64-GB memory. To implement the proposed temporal-GCN and spatial-GCN methods to detect the interaction change of landscape patterns, the Pytorch framework with Pytorch-Geometric (PYG) was used to establish the GCN model, and the labeled training instances were used to train the model.

Data used in the temporal-GCN and spatial-GCN for training and validation were constructed automatically with human interference. For the spatial-GCN dataset, landscape patches were first conceptualized as nodes, namely, the geometric centroid (i.e., nodes) of the land cover patch, and the same land cover type within the one-hop neighbors' landscape unit was calculated with GIS software, as shown in Fig. 1(b) and (c). Second, the line (i.e., edge) between the nodes was connected, and the spatial graph structure was automatically visualized. Finally, RUP, IUP, and SUP were manually selected on the basis of distance and direction between the central and neighbor nodes within the landscape unit of one-hop neighbors. For the temporal-GCN dataset, five types of patch type changes, namely, creation, aggregation, dissection, enlargement, and shrinkage, were collected from land cover maps two consecutive times based on our previous study [12]. The detailed data are presented in Table II.

During the training stage of the GCN model, the Adam optimizer (default parameter) was adopted to update the model until convergence, and the learning rate was set to 0.005. To improve the accuracy of the classification of landscape pattern types, training and test sets were obtained using random sampling. The model was trained using 70% of the data, and the rest of the samples were used for testing. To quantitatively evaluate the performance of temporal-GCN and spatial-GCN,

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TABLE II
DATA USED IN TEMPORAL-GCN AND SPATIAL-GCN FOR TRAINING AND VALIDATION

Model	Sample type	Total	Training	Validation
Spatial-GCN	RUP	430	301	129
	IUP	322	225	97
	SUP	435	305	131
Temporal-GCN	Stable	174	122	52
	Creation	216	151	65
	Enlargement	168	118	50
	Shrinkage	172	120	52
	Aggregation	184	129	55
	Dissection	172	120	52

Notes: RUP denotes a relatively uniform pattern, IUP denotes an incompletely unbalanced pattern, and SUP denotes a severely unbalanced pattern.

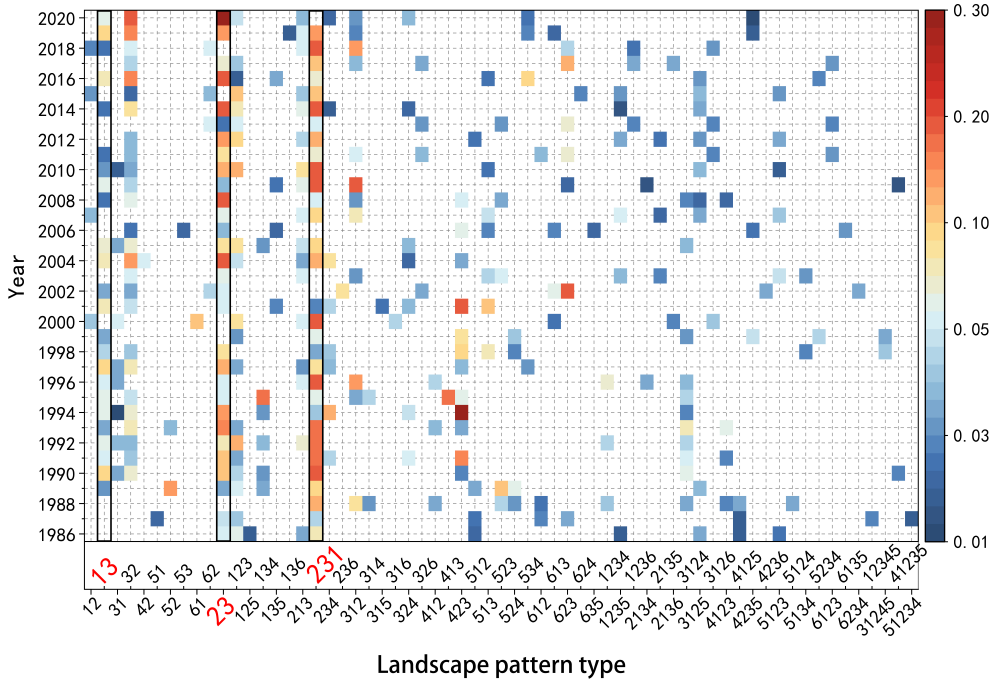

Fig. 8. Percentage of ranked top 10 landscape pattern types in each year from 1986 to 2020. Note: different colors display the percentage of some node occupied overall nodes. Cropland, forest, grassland, water, building land, and unused land are denoted by 1, 2, 3, 4, 5, and 6, respectively. In addition, the first number displays the land cover type of the central node, and the other displays the land cover type of the neighbor's nodes.

the OA, average accuracy (AA), Kappa coefficient (Kappa), and confusion matrix were adopted.

C. Detection of Landscape Pattern Types

In the detection analysis of landscape patterns, the two major steps are generally related to classifying landscape pattern types and selecting dominant landscape pattern types in a 35-year period using spatial-GCN, which better characterizes the change process in the next temporal-GCN.

All available landscape pattern types were calculated to investigate spatiotemporal interactions from 1985 to 2020. To analyze the dominant landscape pattern types across the time period, two factors, namely, the frequency and proportion of landscape pattern types over 35 years, were considered. Fig. 8 shows the ranked top 10 landscape pattern types for each year for 35 years. Three dominant landscape pattern types, forest as central node, grassland, and cropland as neighbor nodes (labeled ⟨231⟩), forest as central node, grassland as neighbor node (labeled ⟨23⟩), cropland as central node, and grassland as neighbor node (labeled ⟨13⟩), stood out according to their frequency and proportion. In other words, grassland and forest coexist with each other, and cropland and grassland likely coexist with each other.

The RUP, IUP, and SUP are shown in Fig. 9. The percentage of landscape pattern types with node attributes for all six land cover classes was distributed throughout the study area [Fig. 9(a)]. The percentage of RUP exhibited a fluctuating increasing trend until the end of the study period, reaching a maximum of 48% in 2000. The percentage of SUP exhibited

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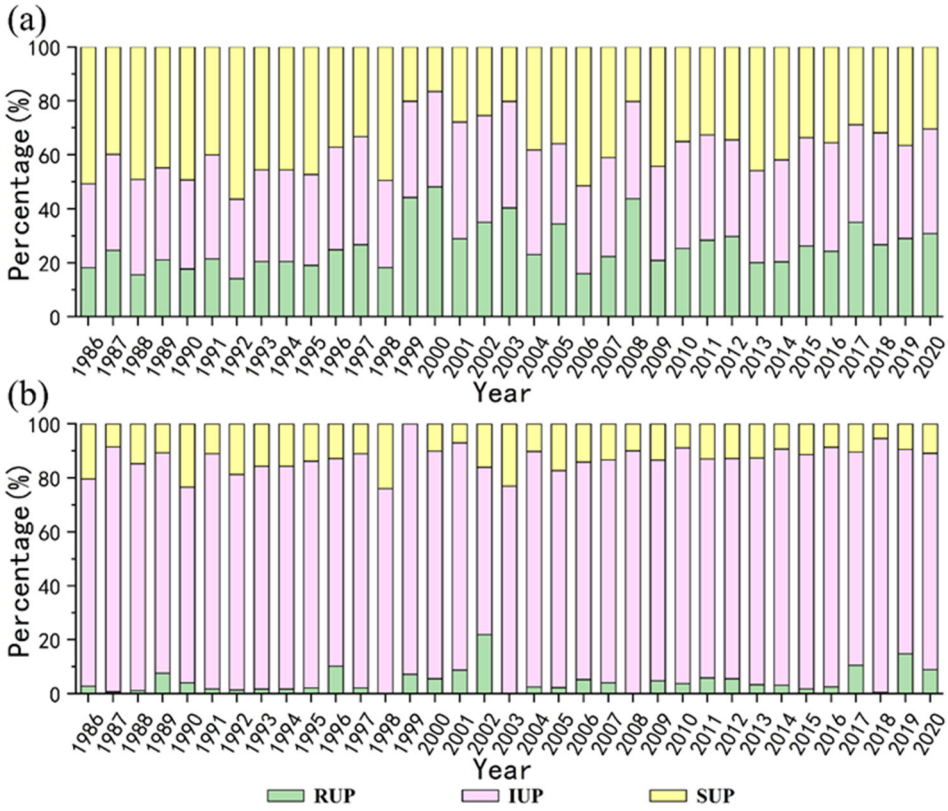

Fig. 9. Percentage of three landscape pattern types in each year from 1986 to 2020. (a) Percentage of RUP, IUP, and SUP with all land cover classes. (b) Percentage of RUP, IUP, and SUP with only forest as the central node, and grassland and cropland as neighbor nodes (labeled ⟨231⟩).

a fluctuating downward trend from 1986 to 2020, reaching a minimum of 16% in 2000. Meanwhile, the percentage of IUP maintained relatively stable from 1986 to 2020, reaching approximately 35%. The percentage of RUP, IUP, and SUP with three dominated node attributes was evaluated (labeled ⟨231⟩, labeled ⟨23⟩, labeled ⟨13⟩), and Fig. 9(b) shows the percentage of landscape pattern types with labeled ⟨231⟩; the higher proportion of labeled ⟨231⟩ belonged to IUP due to large forest patches with a number of small patches of different shapes and sizes, making it the dominant pattern type in the study area. SUP and RUP exhibited subtle decreases and increases, respectively. Notably, landscape pattern types labeled ⟨23⟩ and ⟨13⟩ belonged to an RUP because the central patch was surrounded by the other patches with the same land cover type within the one-hop neighbors' landscape unit.

D. Analysis of Major Landscape Pattern-Type Changes

Based on the above analyses, the dominated landscape pattern types in 35 years were obtained, including forest as the central node, and grassland and cropland as neighbor nodes (hereafter referred to as group ⟨231⟩); forest as the central node and grassland as the neighbor node (hereafter referred to as group ⟨23⟩); and cropland as the central node and grassland as the neighbor node (hereafter referred to as group ⟨13⟩). Therefore, the variation of three landscape pattern types was calculated. Fig. 10 shows the initial appearance time of the three dominant landscape pattern types. To characterize the variation of landscape pattern, the location of the central node displayed the spatial distribution of the landscape units of one-hop neighbors. In addition, the initial appearance times were split into seven segments (i.e., 1986–1989, 1990–1994, 1995–1999, 2000–2004, 2005–2009, 2010–2014, and 2015–2020), and approximately 95% of the initial appearance time of three dominated landscape pattern types occurred in the last 30 years (i.e., 1990–2020). With respect to groups ⟨231⟩, ⟨23⟩, and ⟨13⟩ as landscape pattern types, the initial appearance time mostly occurred in consecutive years (i.e., 2010–2020), two five-year periods (i.e., 1990–1994 and 2015–2020), and ten consecutive years (1990–1999).

The variations in area average landscape metrics, including PC, the distance between two adjacent patches (DP), LPI, and PAF from 1986 to 2020 are shown in Figs. 11–13. These metrics can characterize spatiotemporal variation for the landscape unit of one-hop neighbors. The fitting line was used to verify the trends (i.e., increasing and decreasing) in the time series of landscape metrics from 1986 to 2020. With respect to groups ⟨231⟩ and ⟨23⟩ as landscape pattern types in Figs. 11 and 12, the temporal variation of the area average landscape metrics for group ⟨231⟩ resembled that of group ⟨23⟩. For PC, LPI, and DP, the increasing trends dominated the dynamic change process. For the PAF, decreasing trends dominated the dynamic change process. Based on the above analysis, greater PC, LPI, and DP and less PAF are indicative of a landscape pattern suffering from human activity, such as increasing patches and more regular patch shapes. With

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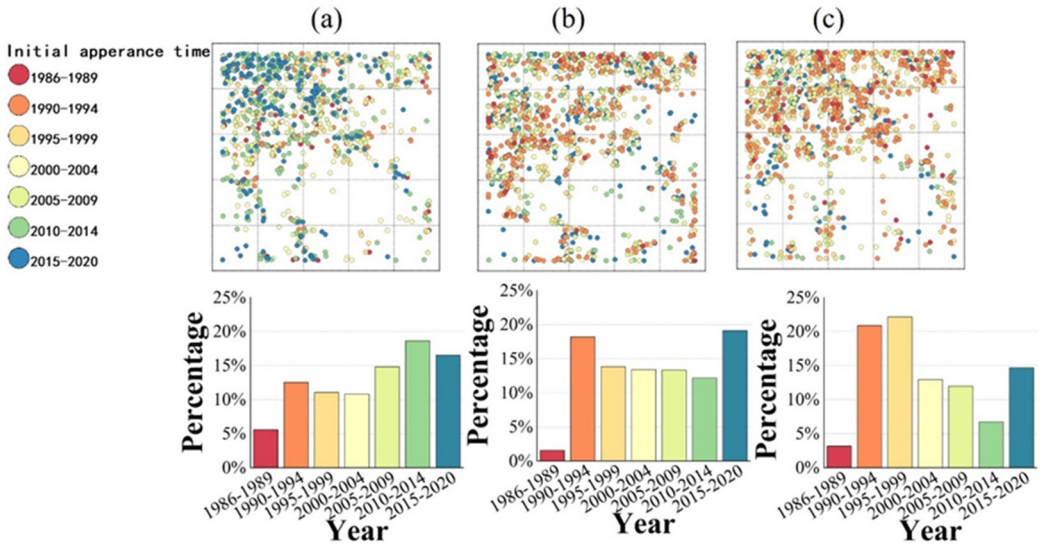

Fig. 10. Spatial distribution for the initial appearance time of dominated landscape pattern types in the study area: (a) forest as the central node, and grassland and cropland as the neighbor nodes (labeled ⟨231⟩), (b) forest as the central node and grassland as the neighbor node (labeled ⟨23⟩), and (c) cropland as the central node and grassland as the neighbor node (labeled ⟨13⟩). Note: location of colored dot denotes the spatial distribution of one-hop neighbors' landscape unit.

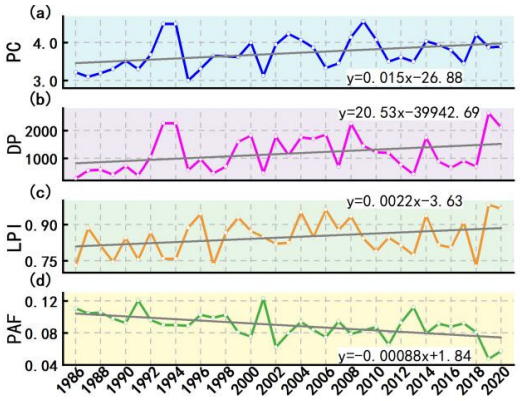

Fig. 11. Area-averaged landscape metrics within one-hop neighbors' landscape unit of forest as the central node, and grassland and cropland as the neighbor nodes (labeled ⟨231⟩) as dominant landscape pattern types from 1986 to 2020: (a) PC, (b) distance of the two adjacent patches (DP), (c) LPI, (d) PAF.

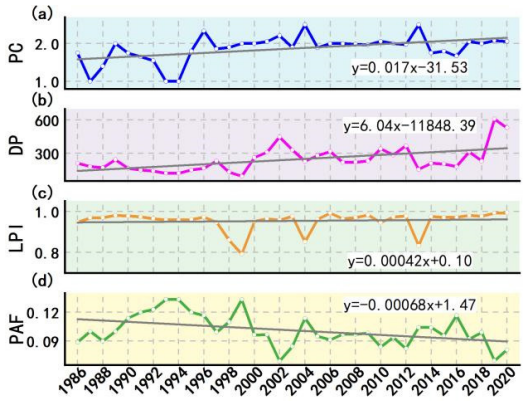

Fig. 12. Area-averaged landscape metrics within one-hop neighbors' landscape unit of forest as the central node and grassland as the neighbor nodes (labeled ⟨23⟩ as dominant landscape pattern types from 1986 to 2020: (a) PC, (b) distance of the two adjacent patches (DP), (c) LPI, and (d) PAF.

respect to group ⟨13⟩, the temporal variation of the area average landscape metrics showed subtle differences between groups ⟨231⟩ and ⟨23⟩. For the LPI and DP, increasing trends dominated the dynamic change process. For PC and PAF, decreasing trends dominated the dynamic change process. Notably, PC in the landscape metrics presented a difference in groups ⟨231⟩, ⟨23⟩, and ⟨13⟩. It was concluded that an increasing number of cropland patches resulted in greater PC, regardless of groups ⟨231⟩ and ⟨23⟩.

E. Reason and Mechanism of Landscape Pattern-Type Changes

Based on the above analysis, single landscape patch change, including creation, aggregation, dissection, enlargement, and shrinkage, were detected by the temporal-GCN. As is well known, landscape pattern-type changes are induced by a single landscape patch; for example, the creation, aggregation, and dissection related to changes in a single landscape patch result in a change in landscape patch types, which leads to changes in landscape pattern types. Enlargement and shrinkage related to changes in a single patch resulted in patch–patch distance and direction, leading to changes in the landscape pattern. To investigate which type of single patch change is the major factor for landscape pattern change, the change in node percentage of the total nodes was calculated. The induction sorting of landscape pattern type changes from 1986 to 2020 in this study area is shown in Fig. 14. As shown in Fig. 14, regardless of groups ⟨231⟩, ⟨23⟩, and ⟨13⟩, creation was the dominant factor resulting in the landscape pattern types. For group ⟨231⟩, the maximum and minimum induction contributions were creation and shrinkage, respectively. For group ⟨23⟩,

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TABLE III
CONFUSION MATRIX OF DETECTION LANDSCAPE PATTERN TYPES BASED ON SPATIAL-GCN

Data	Classification	RUP	IUP	SUP	User's accuracy
Training data	RUP	301	0	0	100%
	IUP	0	225	0	100%
	SUP	0	0	305	100%
	Producer's accuracy	100%	100%	100%	
	Overall accuracy				100%
	Kappa coefficient				1
Validation data	RUP	126	2	1	97.7%
	IUP	1	92	4	94.8%
	SUP	0	5	126	96.2%
	Producer's accuracy	99.2%	92.9%	96.2%	
	Overall accuracy				96.4%
	Kappa coefficient				0.94

Notes: RUP denotes a relatively uniform pattern, IUP denotes an incompletely unbalanced pattern, and SUP denotes a severely unbalanced pattern.

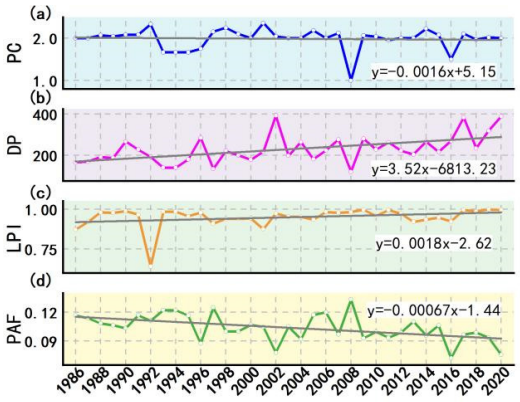

Fig. 13. Area-averaged landscape metrics within one-hop neighbors' landscape unit of cropland as the central node and grassland as the neighbor nodes (labeled ⟨13⟩) as dominant landscape pattern types from 1986 to 2020: (a) PC, (b) distance of the two adjacent patches (DP), (c) LPI, and (d) PAF.

the largest change induction contribution measured was that of creation, followed by enlargement, dissection, shrinkage, and aggregation. For group ⟨13⟩, the change induction contribution decreased in the following order: creation > enlargement > aggregation > shrinkage. This indicates that the change types of a single patch can be responsible for the reason and mechanism of landscape pattern-type change.

F. Detection Performance of Spatial-GCN and Temporal-GCN

The stability of each model was assessed using the slope of the loss curve and the speed of convergence. The convergence of the losses during the training of the spatial-GCN and temporal-GCN is shown in Fig. 15. The NLLLoss function on datasets over 500 epochs is illustrated, with Fig. 15(a) representing the spatial-GCN losses and Fig. 15(b) representing the temporal-GCN losses. We found that the curves from the spatial-GCN and temporal-GCN showed nearly similar variations in their slopes at epoch ranks from 0 to 50. The spatial-GCN and temporal-GCN appeared to converge gradually and showed good convergence behavior, with the loss converging to relatively small values after approximately 200 epochs. Until 500 epochs, the spatial loss values of the GCN and temporal-GCN decreased to 0.049 and 0.128, respectively. Compared with that of the temporal-GCN, the convergence speed of the spatial-GCN was not stable, indicating loss curves fluctuating at the 100th, 250th, and 400th epochs. These results suggest that spatial-GCN and temporal-GCN show promising performance in terms of complex networks.

To assess the credibility of detecting landscape pattern types, a confusion matrix was used to analyze the training and validation sets. First, a comparison of landscape pattern type outputs between spatial-GCN and human visual interpretation was made. The classification accuracy of an RUP, IUP, and SUP are shown in Table III because the patch types relating to land cover class across spatial-GCN have no effect on the accuracy result. For training sets, the landscape pattern types were detected with an OA and Kappa coefficient of 100% and 1, respectively. For validation sets, the landscape pattern types were detected with an OA and Kappa coefficient of 96.4% and 0.94, respectively. Second, a comparison of the patch change type outputs between the temporal-GCN and human visual interpretation was made. The classification accuracies of the six patch change types are listed in Table IV. For training sets, the patch change types were detected with

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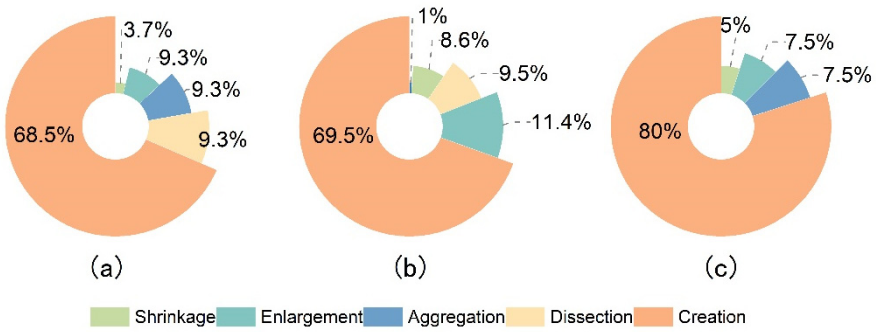

Fig. 14. Induction sorting of dominant landscape pattern type changes from 1986 to 2020 in the study area: (a) forest as the central node, and grassland and cropland as the neighbor nodes (labeled ⟨231⟩), (b) forest as the central node and grassland as the neighbor node (labeled ⟨23⟩), and (c) cropland as the central node and grassland as the neighbor node (labeled ⟨13⟩).

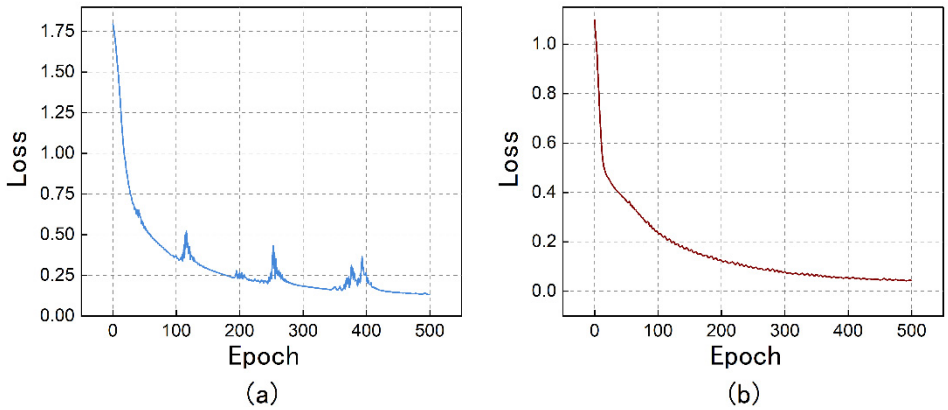

Fig. 15. Loss curve of GCN model in detection landscape pattern types: (a) spatial-GCN model and (b) temporal-GCN model.

an OA and Kappa coefficient of 98.7%, and 0.98, respectively. For validation sets, the patch change types were detected with an OA and Kappa coefficient of 92.0% and 0.90, respectively.

V. Discussion

Remote sensing time series images are important data sources for analyzing the evolution of surface landscape patterns because they contain rich temporal and spatial structure information of the land surface system [4], [5]. Although many change detection algorithms have been developed at the pixel [22], [23], [24], [25], [26] and patch levels [13], [14], they focus on the time-varying characteristics of pixels/objects in images and lack interactive analysis of temporal and spatial structures. Here, our study focused on the change in temporal and spatial structures at the superpatch level by introducing graph theory. Graphs are a type of data structure that models a set of objects (nodes) and their relationships (edges). Previous studies have demonstrated that the graph theory is useful in landscape ecological studies [17]. A GCN may be considered as a combination of graph theory and machine learning algorithms, which can obtain abstract high-level representations by gradually aggregating patch-level features. Therefore, there are two particular advantages for the detection of spatiotemporal variation in landscape patterns based on the GCN. Nodes and edges of GCN are optimal tools that have proven to be effective in representing landscape patches and common boundaries between patches. Alternatively, the GCN can precisely perform a multihop neighbors relationship analysis, such as one- and two-hop neighbors, by evaluating the configuration and connectivity of the landscape.

In our study, an attempt was made to detect landscape pattern evolution using yearly Landsat time series based on the GCN. The node (patch) attribute, distance, and direction were considered to detect landscape pattern types, while four evolution metrics within one-hop neighbors' landscape units, namely, PC, LPI, DP, and PAF, were selected as the metrics of landscape pattern evolution. Our change detection algorithm addressed three aspects of temporal–spatial information: 1) interactions between landscape elements or patches and the linkage of patches, such as group ⟨231⟩, relating to forest grassland cropland as a localized spatial dependency unit; 2) the dominant landscape pattern types in a long time series obtained by capturing the temporal relationships between landscape patches over time; and 3) the temporal dynamics of specific spatial arrangements of patches, classes, or entire landscape mosaics.

We noted which tendency of the GCN approach to analyze the patch change is based on the long time series analysis. The first is how to merge the attributes of nodes and network structure information to achieve the classification of landscape pattern types. The second is how to detect spatiotemporal variation in landscape patterns by mining the possible implicit

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TABLE IV
CONFUSION MATRIX OF DETECTION PATCH CHANGE TYPES BASED ON TEMPORAL-GCN

Data	Classification	Stable	Creation	Enlargement	Shrinkage	Aggregation	Dissection	User's accuracy
Training data	Stable	118	0	0	4	0	0	96.7%
	Creation	0	151	0	0	0	0	100%
	Enlargement	0	2	112	0	4	0	94.9%
	Shrinkage	0	0	0	120	0	0	100%
	Aggregation	0	0	0	0	129	0	100%
	Dissection	0	0	0	0	0	120	100%
	Producer's accuracy	100%	98.7%	100%	96.8%	97.0%	100%	
	Overall accuracy							98.7%
	Kappa coefficient							0.98
Validation data	Stable	49	1	0	2	0	0	94.2%
	Creation	0	61	4	0	0	0	93.8%
	Enlargement	0	1	42	0	6	1	84%
	Shrinkage	0	0	1	51	0	0	98.1%
	Aggregation	0	0	8	0	47	0	85.5%
	Dissection	0	0	0	2	0	50	96.2%
	Producer's accuracy	100%	96.8%	76.4%	92.7%	88.7%	98.0%	
	Overall accuracy							92.0%
	Kappa coefficient							0.90

relationships between nodes and their neighbor nodes through a neural network. Third, the procedure for generating mini-batches should consider the number of layers in the GCN model, as the Kth-order neighborhood for a GCN with K layers must be stored in memory for an exact procedure. Further approximations may be necessary for very large and densely connected graph datasets.

However, further work is required to apply GCN to detect landscape pattern changes on a large scale. First, land cover classes are correctly classified from remote sensing, which has enabled the documentation of detailed landscape pattern information because the misjudgments arising from land cover classes are characterized by fast changes within relatively short time intervals (such as seasonal water). Second, the change in landscape pattern can vary significantly due to the different-hops neighbors' analysis unit, and detecting dominant or focused changes based on the temporal and spatial attention mechanisms on a large scale remains challenging. Third, efficient computing platforms are required to map landscape pattern types and obtain the accurate time and location of landscape patterns, particularly for higher level change features.

In this study, a hybrid spatiotemporal GCN was proposed as a holistic framework to accurately capture both spatial and temporal variations of landscape patterns, which will enrich the theories and methods of remote sensing time series analysis. This approach may offer insight into the complex changing characteristics of the surface landscape structure and order at different temporal and spatial scales, and the complex relationship about human activity matter or energy land use, such as ecosystem services, landscape function, and habits for wildlife.

VI. CONCLUSION

Hybrid-layer GCN was proposed as a holistic framework to accurately capture both spatial and temporal variations of landscape patterns. In contrast to the change detection algorithm at the pixel or patch level and landscape pattern analysis at the whole-region level, the one-hop neighbors' patch was selected as the analysis unit of the GCN network, which provides a fresh perspective for the detection of landscape patterns. Both the node and edge features between nodes were jointly optimized during classification and change detection of the landscape pattern. First, a spatial-GCN module with three parallel graph convolutional layers was adopted to classify landscape pattern types. Three dominant landscape types from 1986 to 2020, namely, forest grassland cropland (group ⟨231⟩), forest grassland (group ⟨23⟩), and cropland grass (group ⟨13⟩), were identified. Second, four evolution metrics, namely, PC, DP, LPI, and PAF, were considered to quantify the interactive changes in landscape pattern types. Third, a temporal-GCN module with two progressive graph convolutional layers was

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used to detect the landscape patch change types (i.e., stable, creation, aggregation, dissection, enlargement, and shrinkage) and could be responsible for the reason and mechanism of landscape pattern change.

Spatial- and temporal-GCN models provide a satisfactory performance in detecting interactive changes in landscape pattern types. They both had better accuracies regardless of the spatial- and temporal-GCN model, with OA $>$ 92% and Kappa coefficient $>$ 0.90. The NLLLoss function over all labeled instances was used as the loss function of the GCN to achieve classification. The loss values of the spatial- and temporal-GCN models were reduced to 0.049 and 0.128, respectively. It is believed that the hybrid GCN model will play an important role in detecting the complex changing characteristics of surface landscape structures in the future.

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